1) A time-harmonic plane wave travelling in free space is described by

$$\vec{H} = 10\, e^{-jx}\, \vec{a}_z \ \text{(mA/m)}. \qquad \vec{a}_k = +\vec{a}_x$$

a) Find the direction of propagation, frequency, phase constant and wavelength.
b) Write the instantaneous expression of the corresponding $\vec{E}$.
c) Calculate the value of the electric field at x =2 m and t =1 ns.
d) Write the expression of the wave time-average power density vector.

a) $\beta = 1$ rad/m ; $v = c$; $\omega = \beta v = 3\times10^8$ rad/s ; $\lambda = 2\pi$ (m)

b) $\vec{E}(x,t) = 10\times10^{-3}(377)\cos(3\times10^8 t - x)\,\vec{a}_y$ (V/m)

c) $\vec{E}(2m, 1ns) = 3.77\cos(0.3-2)\,\vec{a}_y = -0.48\,\vec{a}_y$ (V/m)

d) $\vec{S}_{ave} = \dfrac{(3.77)^2}{2(377)}\,\vec{a}_x = 18.85\,\vec{a}_x$ (mW/m^2)

2) Consider a plane wave propagating in a lossy non-magnetic medium. The impedance of this medium is $\eta = 200 + j90$ Ohms . Given the expression for the electric field as,

$$\vec{E} = \sqrt{2}\, e^{\alpha y}\cos(2\pi\times10^9 t + \beta y - \tfrac{\pi}{4})\,\vec{a}_z \ \text{(V/m)}$$

a) Find the medium dielectric constant.
b) Find the medium conductivity.
c) Find the wave phase constant.
d) Find the wave attenuation constant.
e) Find the wavelength in the direction of propagation.
f) Write the corresponding $\vec{H}$ in phasor form.
g) What percentage of the time-average power density magnitude will the wave lose for every centimeter of propagation inside this medium?

a) $\eta = 219\angle 24.2^\circ\ (\Omega)$; $\dfrac{\sigma}{\omega\epsilon} = \tan(2\times24.2) = 1.13$

$$|\eta| = 219 = \frac{\eta_0/\sqrt{\epsilon_r}}{\left[1+\left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/4}} \Rightarrow \epsilon_r = 2.0$$

b) $\dfrac{\sigma}{\omega\epsilon} = 1.13 \Rightarrow \sigma = 1.13\times2\pi\times10^9\times2\times8.85\times10^{-12} = 0.126$ (S/m)

c) $\beta = 2\pi\times10^9\sqrt{\dfrac{4\pi\times10^{-7}\times2\times8.85\times10^{-12}}{2}\left[\sqrt{1+1.13^2}+1\right]} \approx 33$ rad/m

d) $\alpha \approx 15$ Np/m

e) $\lambda = 2\pi/\beta = 0.19$ (m)

f) $\vec{H} = \dfrac{\sqrt{2}}{219}\, e^{15y}\, e^{-j\pi/4}\, e^{-j24.2^\circ}\, e^{j33y}\,(-\vec{a}_x)$ A/m

g) Every 1 cm the decay in S_{ave} is $e^{-2(15)(0.01)}$. The loss is about 26%.